Optional Advanced Step

Enabling data=journal in cmdline.txt - for experienced users only

WARNING - this step can stop the Raspberry Pi from booting

- A mistake in cmdline.txt (for example a second line, or a typo) can leave the Pi unable to start. A Pi that cannot start cannot be fixed over SSH or the network.
- Do **not** enable this on a unit you cannot physically reach, and do not enable it without an image backup of the SD card.
- The standard installation (document 1) already protects the data files. **You do not need this step.**

1. What it does and why it is optional

Adding `rootflags=data=journal,commit=1` to cmdline.txt makes the ext4 root filesystem write file *data* (not only metadata) through its journal and flush every second. After a power cut the journal is replayed at boot. It is an extra safety layer on top of the atomic writes from document 1. Because atomic writes already stop files from being left empty or half-written, the additional benefit is small, while the risk is a boot failure.

2. Do not enable it if...

- you cannot physically reach the Raspberry Pi (remote cabinet, no one on site);
- you have no image backup of the SD card and no spare SD card;
- you are not comfortable editing files on the boot partition from a PC;
- the unit is already working correctly after the standard installation.

3. Before you enable it - checklist

- Standard installation (document 1) is done and verified.
- You have an image backup of the SD card (or a spare card with the same image).
- You have a PC with an SD card reader and can open the boot partition (called bootfs).
- Someone with physical access is available if the Pi does not start.

4. How to enable it

```
cd /home/alice/broker
sudo ENABLE_ROOTFLAGS=1 ./harden_sd.sh
```

Then check that cmdline.txt is still **one single line**:

```
wc -l /boot/firmware/cmdline.txt     # must print 1 (or 0)
cat /boot/firmware/cmdline.txt
```

The line must end with `rootflags=data=journal,commit=1`, separated from the previous option by a single space, with no second line. On older systems the file is /boot/cmdline.txt. A backup exists as cmdline.txt.bak.

If it looks correct, reboot, then verify:

```
sudo reboot
# after boot:
grep ' / ' /proc/mounts     # the options should include data=journal
```

5. If the Pi does not start after enabling it

Step	Action
1	Power off. Remove the SD card and put it in a PC (the boot partition appears as a drive called bootfs).
2	Open **cmdline.txt** with a plain text editor (Notepad, TextEdit in plain-text mode, nano).
3	Either replace its content with the content of **cmdline.txt.bak**, or delete only `rootflags=data=journal,commit=1` (and the space before it). The file must remain one single line.
4	Save, safely eject, put the card back in the Pi and power on.
5	If it still does not start, write the image backup back to the SD card (Raspberry Pi Imager or Win32DiskImager).

Do not use a word processor (Word, WordPad) - it can add hidden characters or line breaks.

6. How to turn it off again (when the Pi still boots)

```
sudo cp /boot/firmware/cmdline.txt.bak /boot/firmware/cmdline.txt
sudo reboot
```

If the .bak file is missing, remove the text `rootflags=data=journal,commit=1` from the line with `sudo nano /boot/firmware/cmdline.txt` and keep it as a single line.

7. Troubleshooting

Symptom	Likely cause	What to do
Pi does not start, no network	cmdline.txt broken (extra line, typo)	Fix or restore cmdline.txt from a PC (section 5)
Boots, but data=journal not in /proc/mounts	Option not applied or an older file path	Check the file path and that the line was saved; reboot
Boots into emergency / read-only mode	Filesystem error or conflicting root options	Restore cmdline.txt.bak; if it persists, restore the image backup